

Return Status

- Famous helloworld.c example revisited

```
#include <stdio.h>
int main()
{
    printf("helloworld :-)\n");
    printf("hello OS\n");

    return 0;
}
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus$ gcc -o helloworld helloworld.c
lih@lih:~/SF/OS_Demo/ReturnStatus$ ./helloworld ; echo $?
helloworld :-)
hello OS
0
```

```
#include <stdio.h>
int main()
{
    printf("helloworld :-)\n");
    printf("hello OS\n");

    return 1;
}
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus$ ./helloworld; echo $?
helloworld :-)
hello OS
1
```

Return Status

- Indispensable role of the return status

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive .  
.:  
total 1  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c  
drwxrwx--- 1 root vboxsf 0 9月 24 19:38 rm_dir  
  
./rm_dir:  
total 24  
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:37 helloworld  
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:37 helloworld_exit1
```

Return Status

- Indispensable role of the return status

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive .  
.:  
total 1  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c  
drwxrwx--- 1 root vboxsf 0 9月 24 19:38 rm_dir  
  
./rm_dir:  
total 24  
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:37 helloworld  
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:37 helloworld_exit1
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ cd rm_dir ; rm * ; cd ..  
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive  
.:  
total 1  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c  
drwxrwx--- 1 root vboxsf 0 9月 24 19:46 rm_dir  
  
./rm_dir:  
total 0
```

Return Status

- Indispensable role of the return status

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive .
.:
total 1
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c
drwxrwx--- 1 root vboxsf 0 9月 24 19:38 rm_dir

./rm_dir:
total 24
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:37 helloworld
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:37 helloworld_exit1
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ cd rm_dir_utopia ; rm * ; cd ..
bash: cd: rm_dir_utopia: No such file or directory
rm: cannot remove 'rm_dir': Is a directory
lih@lih:~/SF/OS_Demo/ReturnStatus$ cd T
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive
.:
total 0
drwxrwx--- 1 root vboxsf 0 9月 24 19:52 rm_dir

./rm_dir:
total 24
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:52 helloworld
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:52 helloworld_exit1
```

Return Status

- Check the return status before execution

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive .  
.:  
total 1  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c  
drwxrwx--- 1 root vboxsf 0 9月 24 19:38 rm_dir  
  
./rm_dir:  
total 24  
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:37 helloworld  
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:37 helloworld_exit1
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ cd rm_dir_utopia ; echo $?  
bash: cd: rm_dir_utopia: No such file or directory
```

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Return Status

- Check the return status before execution

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive .
.:
total 1
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c
drwxrwx--- 1 root vboxsf 0 9月 24 19:38 rm_dir

./rm_dir:
total 24
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:37 helloworld
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:37 helloworld_exit1
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ cd rm_dir_utopia ; if [ $? = 0 ]; then rm * ; cd ..; fi
bash: cd: rm_dir_utopia: No such file or directory
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive
.:
total 1
-rwxrwx--- 1 root vboxsf 99 9月 24 20:02 helloworld.c
-rwxrwx--- 1 root vboxsf 99 9月 24 20:02 helloworld_exit1.c
drwxrwx--- 1 root vboxsf 0 9月 24 19:52 rm_dir

./rm_dir:
total 24
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:52 helloworld
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:52 helloworld_exit1
```

Shell Script

- Group commands into a shell script

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ ls -l --recursive .  
.:  
total 1  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld.c  
-rwxrwx--- 1 root vboxsf 99 9月 24 19:37 helloworld_exit1.c  
drwxrwx--- 1 root vboxsf 0 9月 24 19:38 rm_dir  
  
./rm_dir:  
total 24  
-rwxrwx--- 1 root vboxsf 8304 9月 24 19:37 helloworld  
-rwxrwx--- 1 root vboxsf 8312 9月 24 19:37 helloworld_exit1
```

```
lih@lih:~/SF/OS_Demo/ReturnStatus/T$ vi rm_dir_utopia.sh
```

```
cd rm_dir_utopia  
if [ $? = 0 ]; then  
    rm *  
    cd ..  
fi
```